FINANCE / BANK MARKETING

Bank Term Deposit Campaign Intelligence

A Tableau and Python capstone on how customer segments, contact history, and campaign design shape subscription lift.

11.70%

Baseline term-deposit subscription rate

Team ID Sec-B G-2

Members Harshvardhan Gupta, Kartik Madaan, Dev Tyagi
Shitanshu Tiwari, Satwik Mani Tripathi, Siddharth Dangi

Project signal

5,289

subscriptions out of 45,211 contacts

23.07%

previously contacted clients convert nearly 2x better

UCI Bank Marketing dataset | Finance sector | Executive analytics deck

Context and problem statement

The bank needs to sell term deposits, but the campaign budget is wasted if every lead is treated the same way.

Business context

Retail banking campaigns are high-volume and low-conversion. In this dataset, the bank contacted 45,211 clients and only 11.7% subscribed. That means the core challenge is not reaching more people, but identifying which people deserve the next call.

Decision to unlock

Which customers should be contacted first, which channels should be prioritized, and when should the bank stop repeating low-yield calls?

What the analysis must answer

1) Which demographic and financial segments respond best? 2) Does prior contact history matter more than raw volume? 3) How can the bank lift subscriptions without inflating call cost or annoying low-propensity clients?

Data engineering

The raw UCI file was cleaned into a Tableau-ready dataset with derived segments for targeting and reporting.

Source

UCI Bank Marketing CSV
45,211 rows | 17 raw fields

Clean

Standardized labels, preserved unknowns, converted yes/no response to binary y

Derive

age_group, balance_band, campaign_band, previously_contacted, duration_minutes

Publish

Processed dataset with 22 columns for analysis and Tableau

Source and size

Raw source: UCI Machine Learning Repository.
Rows: 45,211.
Columns after cleaning: 22.
Granularity: one row per client contact record.

Cleaning steps

Stripped whitespace, kept unknown categories explicit, re-coded response into binary y, derived age and balance bands, and created a prior-contact flag so Tableau filters could stay business-readable.

Data dictionary snapshot

Field	Meaning	Role
job	Occupation category	Segmentation
balance	Avg. yearly balance	Financial context
previous	Prior contacts count	Campaign history
poutcome	Previous outcome	Root cause
y	Subscribed? (0/1)	North-star KPI

KPI and metrics framework

The deck tracks outcome, targeting, and financial context metrics so the bank can judge both performance and risk.

Outcome KPIs

- **Subscription rate**
  11.70% of clients subscribed.
- **Total subscriptions**
  5,289 wins from 45,211 contacts.

Targeting KPIs

- **Avg contacts / client**
  2.76 touches per client.
- **Previously contacted rate**
  18.3% had prior campaign exposure.
- **Best channel conversion**
  Cellular outperforms telephone.

Financial context KPIs

- **Avg yearly balance**
  ■1,362 overall.
- **Default rate**
  1.80% of clients.
- **Loan rate**
  16.02% of clients.

Framework logic

Outcome KPIs tell us whether the campaign worked. Targeting KPIs show how the bank is spending attention. Financial context KPIs explain which customers are safe, risky, or simply more likely to convert.

Key insights from EDA

The analysis reveals who responds, when they respond, and which segment characteristics deserve a call first.

1. **Students and retirees convert best.** Their subscription rates are materially above the project average, which makes them high-priority prospect pools.

2. **Single clients respond better than married clients.** Marital status is not the target on its own, but it helps refine message framing and timing.

3. **Higher balances correlate with stronger conversion.** Medium and high balance bands outperform low and zero/negative balances.

4. **Cellular contact beats telephone.** Unknown or untracked channels underperform sharply, suggesting poor routing quality.

5. **Seasonality matters.** March, September, October, and December have the strongest response rates, so campaign timing is not random.

Segment heatmap

Conversion rate by age group and balance band

	negative_or_zero	low	medium	high
18-24	10.9%	27.5%	28.0%	28.6%
25-34	7.2%	12.2%	15.9%	16.6%
35-44	5.9%	8.6%	13.5%	13.1%
45-54	6.4%	8.5%	11.5%	12.3%
55-64	7.8%	11.0%	17.2%	17.6%
65+	32.2%	43.8%	44.0%	36.3%

Low High

Business readout

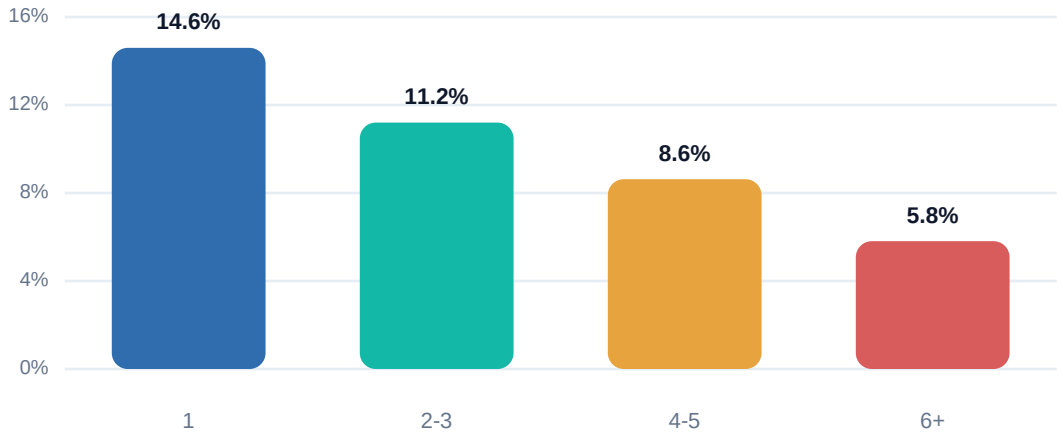
The strongest segments are old, warm, and financially healthier. The weakest are low-balance clients and low-context channel records.

Advanced analysis

A simple root-cause view shows that warm leads and low-touch campaigns outperform broad repeated calling.

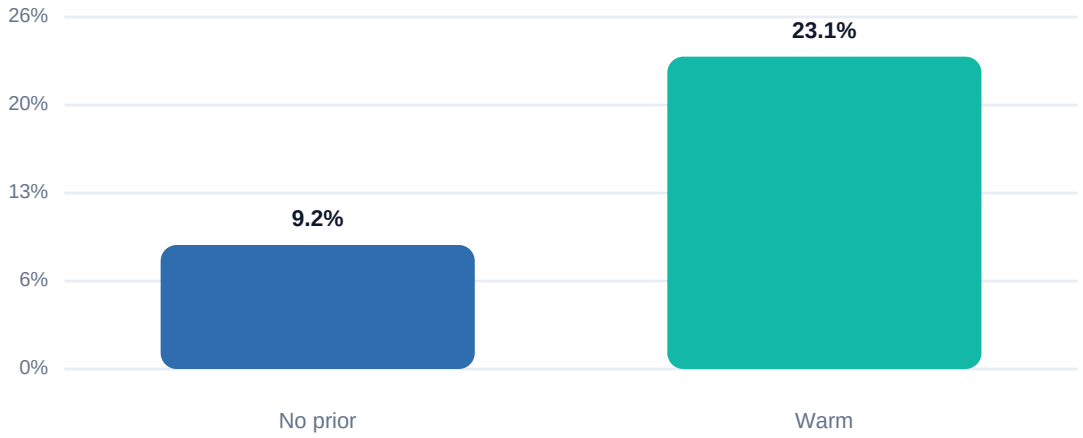
More contacts, lower conversion

Subscription rate by campaign band



Warm leads convert much better

Previously contacted vs first-time contact



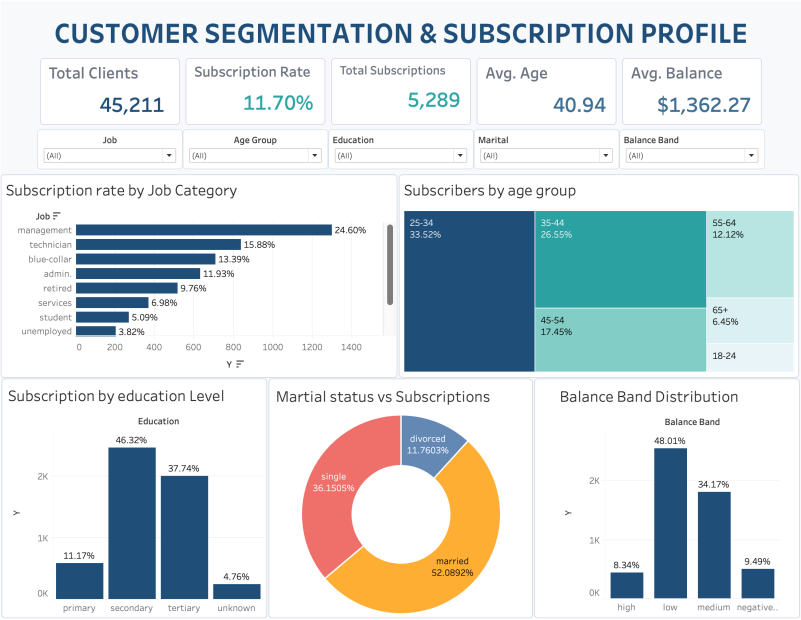
Interpretation The problem is not only volume. It is misallocated volume: too many low-propensity clients receive repeated outreach. That makes segmentation, contact caps, and warm-lead routing the main optimization levers.

Dashboard walkthrough

The executive view summarizes what happened; the operational views explain why and where the lift lives.

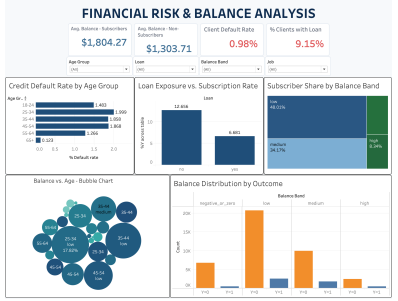
Executive view

Dashboard 1: customer segments, balance bands, and overall conversion



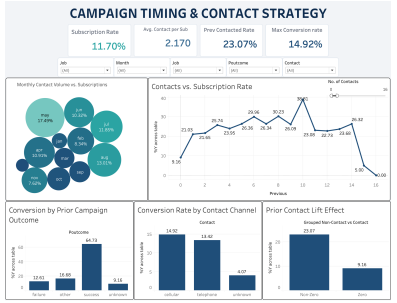
Operational view 2

Dashboard 3: risk lens



Operational view 1

Dashboard 2: contact lens



How to read the dashboards

Use the executive view for leadership reporting. Use the operational views to filter by job, age group, education, contact type, balance band, and prior campaign history before deciding who should be called next.

Recommendations

Each recommendation is directly tied to one of the strongest observed signals in the analysis.

Prioritize students, retirees, and 65+ clients

These groups consistently show the highest conversion, so they should enter the call queue first.

Use prior-contact flags and previous outcomes

Warm leads and prior success history deliver the biggest lift, so the CRM should route them differently.

Cap repeated outreach after 3 touches

Conversion drops sharply after that point, so over-contacting should be treated as waste.

Prefer cellular-first contact plans

The cellular channel beats telephone, while unknown channels are weak and should be cleaned up.

Impact and value

The opportunity is to convert the same outreach volume into a better subscription yield and fewer wasted calls.

Baseline

11.7%

1,170 subscriptions per 10,000 contacts

Previously contacted

23.1%

2,307 subscriptions per 10,000 contacts

Prior success

64.7%

6,473 subscriptions per 10,000 contacts

Illustrative value model

Assumption used only for presentation: each incremental subscription contributes ■50 of net value.

If 10,000 contacts are routed to previously contacted leads:

Baseline = 11.7% -> 1,170 subscriptions

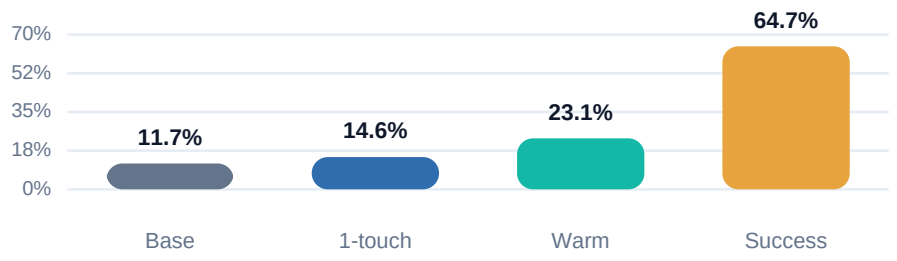
Targeted = 23.1% -> 2,307 subscriptions

Incremental lift = 1,137 extra subscriptions

Illustrative value = ~■56,864 per 10,000 contacts

Scenario lift

Estimated response scenarios



Limitations and next steps

The dashboard is decision-ready, but the next layer should optimize what to do with each lead, not just describe who converted.

Limitations

1) No channel-cost data, so ROI is estimated rather than exact.
2) `duration` is useful for analysis but leaks outcome information if used in prediction.
3) The dataset does not contain product price, interest-rate, or offer-level detail.
4) The campaign is a single historical window, so seasonality should be validated in new data.

Next steps

1) Build an uplift model or scorecard to rank leads before dialing.
2) Test contact caps with an A/B experiment.
3) Combine this bank dataset with call-cost and margin data for a true ROI model.
4) Expand the dashboard with monthly trend overlays and segment-level drilldowns.

Closing note

The project already tells the bank where the lift is. The next step is to turn those signals into a routing engine.

In other words: better targeting, fewer wasted calls, and a cleaner path from Tableau insight to campaign action.